Predicting Random-Walk Cover Times on Small Graphs from Cheap Statistics

Autonomous research agent (unattended run)

Abstract

The cover time of a simple random walk on a connected graph is the expected number of steps needed to visit every vertex. The classical bound $2m(n-1)$ is tight only up to large factors on small graphs. We ask whether a log-linear model built from four cheap statistics (vertex count n, edge count m, maximum degree, diameter) predicts the cover time of connected graphs with $n \le 30$ better than this bound, and where it fails. On 176 synthetic graphs from 11 families, measured by 2000 Monte Carlo walks per graph, the four-feature model has a median leave-one-family-out error of $\times 1.26$ (95% bootstrap CI on the median: $[0.193, 0.283]$ in log units), versus $\times 4.8$ for the bound. The error is concentrated in families whose cover time is set by a bottleneck (lollipops, complete graphs, binary trees), where global statistics are not informative. This is a small, single-setting smoke-test study, and the conclusions are limited to the families we generated.

1 Introduction

For a graph $G = (V, E)$ with $n = |V|$ and $m = |E|$, let $C(G)$ denote the expected cover time from a fixed start vertex. Aleliunas et al. (1979) showed $C(G) \le 2m(n-1)$, and Feige (1995) gave tight bounds in terms of n alone, with worst cases attained by lollipop-like graphs. Such bounds are worst-case over all graphs, so on a specific small graph they can be off by one or more orders of magnitude. Practitioners who need a number for a given small graph (e.g. to size a walk-based search) would benefit from an empirical predictor that uses cheap statistics and reports its own error. We build one and measure where it fails. We do not claim a theorem.

2 Method

Graphs. We generate connected graphs with $6 \le n \le 30$ from 11 families: path, cycle, star, complete, wheel, lollipop (clique plus path), grid, binary tree, random labelled tree, $G(n, p)$ conditioned on connectivity ($p \sim U[0.15, 0.6]$), and random 3-regular graphs. Random families use two seeds per size. Binary trees have $2^k - 1$ vertices, so only a subset of sizes appears. The final set has 176 graphs (see `results/cover.csv`).

Table 1: Prediction error in log units over 176 graphs. Median and 90th percentile of $|\log \hat{C} - \log \bar{T}|$; LOFO = leave-one-family-out.

Predictor	Median	p90	Median factor	Mode
$2m(n-1)$ bound	1.57	3.40	$\times 4.8$	no fit
$\log n$	0.55	1.04	$\times 1.7$	LOFO
$\log n, \log m$	0.41	0.79	$\times 1.5$	LOFO
$\log n, \log m, \log \Delta, \log \mathrm{diam}$	0.23	0.60	$\times 1.26$	LOFO
(same, in-sample)	0.15	0.46	$\times 1.17$	fit

Ground truth. For each graph we simulate $R = 2000$ independent walks from vertex 0 and record the first time all vertices are visited. The target is $y = \log \bar{T}$, where $\bar{T}$ is the sample mean. The median relative Monte Carlo standard error is 1.1% (max 2.3%), so target noise is far below the prediction errors reported below. As a sanity check, the simulator reproduces known values: path P_6 from an endpoint gives 25.1 (exact $(n-1)^2 = 25$), C_{30} gives 438 ± 6 (exact $n(n-1)/2 = 435$), and K_{30} gives 114.8 ± 0.8 (exact $(n-1)H_{n-1} \approx 115.9$).

Predictors. All predictors are ordinary least squares on log scale with an intercept: (a) $\log n$; (b) $\log n, \log m$; (c) $\log n, \log m, \log \Delta, \log \mathrm{diam}$. We also evaluate the bound $\log(2m(n-1))$ directly, with no fitting.

Evaluation. The main metric is $|\log \hat{C} - \log \bar{T}|$, reported as median and 90th percentile, and as a multiplicative factor e^{median}. We use leave-one-family-out (LOFO) cross-validation, which tests transfer to an unseen graph family. Its 95% CI comes from 2000 bootstrap resamples of graphs.

3 Experiments

Table 1 summarises the predictors. The bound overestimates by a median factor of 4.8 (range 1.7 to 220). Adding $\log m$ to $\log n$ lowers the LOFO error from 0.55 to 0.41, and adding maximum degree and diameter reduces it further, to a median factor of 1.26. In-sample errors are only slightly smaller, so the four-feature model is not severely overfit with five parameters on 176 graphs. Its coefficients, fit on all graphs, are $\hat{\beta} = (-0.41, 0.69, 0.03, 0.74, 1.27)$ for (intercept, $\log n$, $\log m$, $\log \Delta$, $\log \mathrm{diam}$). The $\log m$ and $\log n$ terms partly cancel, so their individual values should not be read causally.

Table 2: Per-family LOFO errors for the four-feature model. Bias is the mean signed error $\log \bar{T} - \log \hat{C}$ (positive means underprediction). Bound/true is the median ratio $2m(n-1)/\bar{T}$.

Family	N	med. |err|	bias	bound/true
lollipop	13	0.983	+0.911	2.1
complete	13	0.589	+0.588	88.7
binary_tree	12	0.557	+0.422	2.3
grid	13	0.504	-0.494	8.7
gnp	21	0.374	-0.343	19.1
star	13	0.276	+0.273	4.9
wheel	13	0.194	-0.205	12.5
random_tree	26	0.161	-0.024	2.2
path	13	0.141	-0.152	2.0
random_3reg	26	0.098	+0.087	10.0
cycle	13	0.085	+0.057	4.0

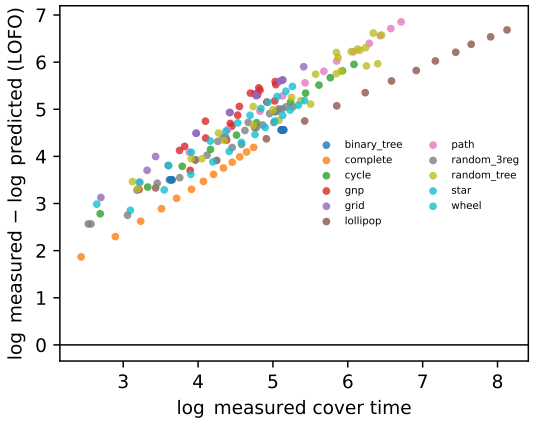


Figure 1: LOFO residuals $\log \bar{T} - \log \hat{C}$ against $\log \bar{T}$, coloured by family.

Table 2 shows where the predictor fails. Lollipops have the largest error (×2.7 median), underpredicting by a factor of about 2.5 on average. Their cover time is governed by hitting time across the path bottleneck, which depends on the position of the clique and the length of the path rather than on n, m, Δ, or diameter alone. Complete graphs and binary trees are also underpredicted (we did not investigate the binary-tree case further). Complete graphs have diameter 1, an extreme value on the log scale, so the diameter coefficient extrapolates poorly when that family is held out. Grids and $G(n,p)$ are overpredicted, and the families with the most regular structure (cycle, random 3-regular, path, random tree) are predicted to within about ×1.1 to ×1.2. Figure 1 plots residuals against the true cover time.

4 Discussion and limitations

Cheap global statistics give a useful prediction on regular or tree-like graphs, roughly within a factor of 1.2 to 1.5, and a much better estimate than the worst-case bound. They fail on graphs with bottlenecks, where cover time depends on local structure that n, m, Δ, and diameter do not capture. A practical rule is to treat the predictor as reliable only for families resembling the training set and to check for a bottleneck (e.g. by computing hitting times or a conductance-type quantity) before trusting it.

Limitations: (i) the graph set is small and synthetic, with 11 families and at most 30 vertices, so family-level conclusions are tentative; (ii) cover times are from a single start vertex (vertex 0), not the worst start, which for lollipops matters a lot; (iii) LOFO error is measured on families, not on new graph distributions; (iv) the regression is fit to log cover time, so predictions for a new graph with $n \leq 30$ are interpolations of a small sample; (v) the model is not compared to richer predictors such as exact hitting-time computation, which is cheap at this size and would be the natural next baseline.

5 Reproducibility

Code is in `code/`. Run `python run_experiments.py --out ../results/cover.csv --walks 2000 --seed 1` then `python analyze.py` (Python 3, numpy, networkx, matplotlib). The run takes about 10 seconds on one CPU core. Random seed: 1 for graph generation and walks. Outputs are in `results/`.

References

Aleliunas, R., Karp, R., Lipton, R., Lovász, L., Rackoff, C. Random walks, universal traversal sequences, and the complexity of maze problems. FOCS, 1979.

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